Tagnon Okoumassoun

+1 (303) 847-9997 | tagnonprivael@gmail.com | LinkedIn | Github | Website

RESEARCH INTERESTS

MS/PhD in Robotics and Autonomous Systems. Focus on Multi-robot systems coordination, joint prediction and planning, and multi-agent reinforcement learning (MARL).

EDUCATION

Master of Engineering, Electrical and Computer Engineering — *University of Ottawa, Canada | 2021–2023*

- Supervised by Dr. Ismaeel Al Ridhawi. Research on federated learning and distributed systems for UAV swarms.

Bachelor of Engineering, Electronics and Communications — *All Nations University College, Ghana | 2016–2020*

- GPA: 3.73/4.00. Ranked 1st of 55 graduates (Best Graduating Student)
- Courses: Physics, Control Systems, Embedded Systems, Microprocessors, Communication Protocols

RESEARCH EXPERIENCE

Risk-aware-sensor-placement — Independent work (Baseline: Kim et al., IEEE SysCon 2025) | Github Repo | 2026

- Demonstrated that void probability is convex in sensor capability in a multi-sensor placement task.
- Derived the Jensen gap to first order in capability variance and verified the prediction in simulation.
- Formulated a risk-aware planner with a mean standard deviation objective; in simulation it holds 5th-percentile void probability flat under growing uncertainty while the nominal planner's degrades.

PROJECTS

pursuer-evader — Multi-Agent Reinforcement Learning, JaxMARL | Github Repo | 2026, In Progress

- Cooperative pursuit-evasion: three pursuers, one evader; decentralized pursuer policies trained with IPPO.
- Evader drawn from $K = 3$ parameterized policy classes.

ar4-color-sorting-arm — 6-DOF Color sorting arm, Collaborative Project | Github Repo | 2026, In Progress

- CAD-designed arm, Gazebo simulation, MoveIt 2 motion planning, RGB-D color detection.
- Contribution: designed the gripper base in CAD; built the URDF assembly and integrated it into Gazebo; designed the final simulation world with boxes and bins.

adibot — Autonomous object-following robot simulation | Github Repo | 2026

- Differential-drive robot follows a moving object through a cluttered world: constant-velocity target prediction, A* global planning on an occupancy grid, pure-pursuit tracking, fixed-rate replanning.

air-ground-ops — Multi-Robot Task Allocation (ROS 2 / Gazebo) | Github Repo | 2026

- Heterogeneous team testbed: UAV provides overhead sensing and publishes a coarse occupancy estimate; three UGVs execute assigned navigation tasks.
- Implemented the Hungarian algorithm as a centralized optimization-based allocation baseline across multiple target locations.

INDEPENDENT COURSEWORK

CS223A: Introduction to Robotics (June 2026) — O. Khatib, Stanford Engineering Everywhere

- Forward/inverse kinematics, Jacobians, dynamics, control.

Modern Robotics: Foundations of Robot Motion (April 2026) – Kevin M. Lynch, Northwestern University, Coursera

- Degrees of Freedom, Configuration Space, Rigid-Body Motions

AWARDS & HONORS

- **Best Company Value Idea (top 6 of 85)** – iCon Hackathon, Honda R&D Americas (2026)
- **Best Conference Paper Award**, IEEE Cyber Science Congress (2023)
- **Best Graduating and Outstanding Student**, School of Engineering, All Nations University College (2020)

PUBLICATIONS

- [1] Tagnon P. Okoumassoun, Ismaeel Al Ridhawi, et al. “Blockchain-Enabled SAGIN Communication for Disaster Prediction and Management.” IEEE CyberSci, 2023. [Link]
- [2] Tagnon P. Okoumassoun, Anita Antwiwaa, et al. “Performance Evaluation of Optical Amplifiers in a Hybrid RoF-WDM Communication System.” Journal of Communications, vol. 18, no. 8, 2023. [Link]
- [3] Tagnon P. Okoumassoun, Anita Antwiwaa, et al. “Free Space Optics Link Performance Estimation Under Diverse Conditions.” Journal of Communications, 2022. [Link]

SKILLS

- **Programming:** Python, C++, MATLAB
- **Robotics & Simulation:** ROS 2, Gazebo Sim, MuJoCo, URDF/Xacro, RViz
- **AI & Computer Vision:** Pytorch, OpenCV, Jax, CUDA
- **Tools:** Linux, shell scripting, Git, Docker

WORK EXPERIENCE

Software Test Engineer – Honda R&D Americas (Contract)

October 2023 – Present | Raymond, OH (USA)

- Automated test development for embedded infotainment software on hardware-in-the-loop benches, covering CAN, automotive Ethernet, ADB, and OTA interfaces.
- Built a Python-based test assistant that converts manual test sequences into validated automation; approx. 30% reduction in manual test effort.

AFFILIATIONS

- Black in Robotics (BiR) – Professional Member, 2026
- Emerging Leaders in AI (ELAI) – Graduate Mentee, 2026-2027